Internship Presentation

Rakrish Dhakal

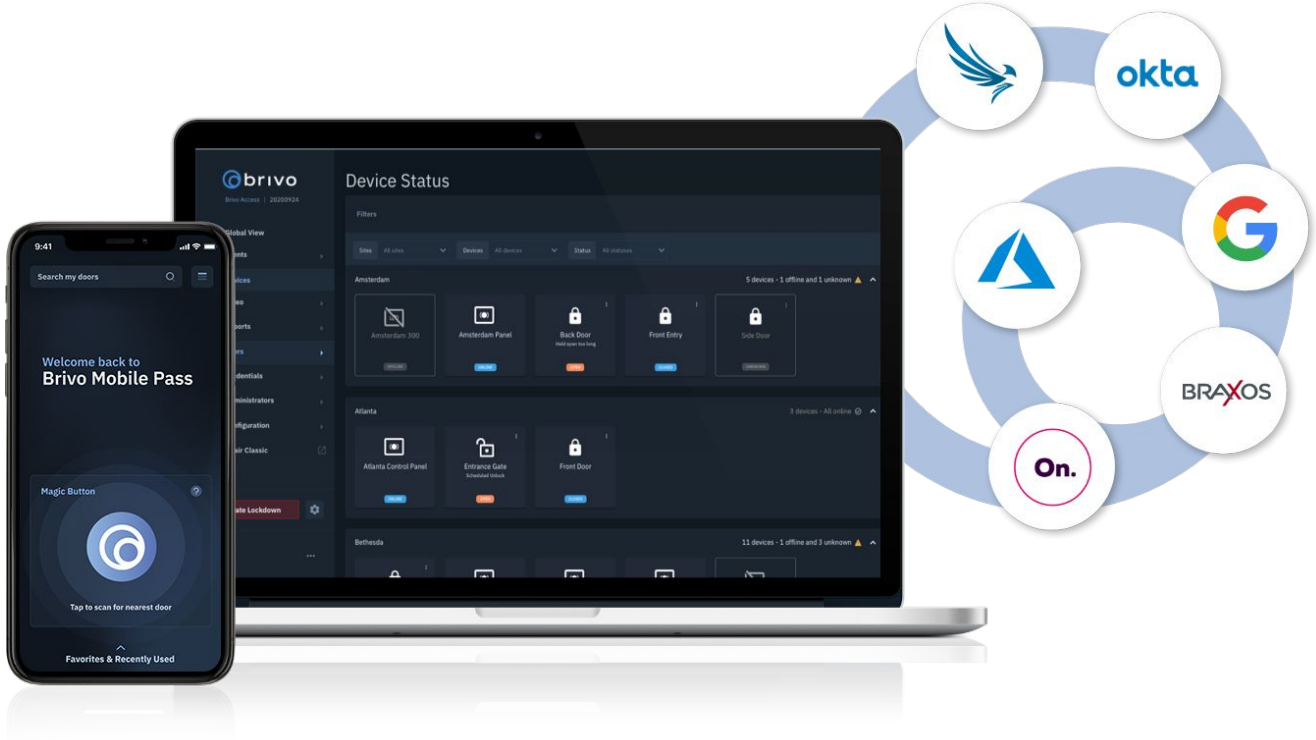
May contain engineering secrets so please don’t record :)

Brivo

- SaaS provider for physical access control

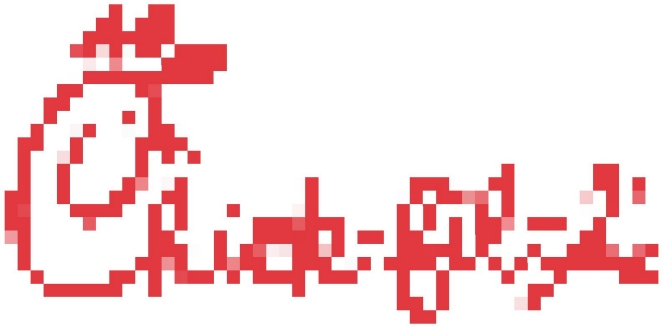
Flagship Products

Flagship Products



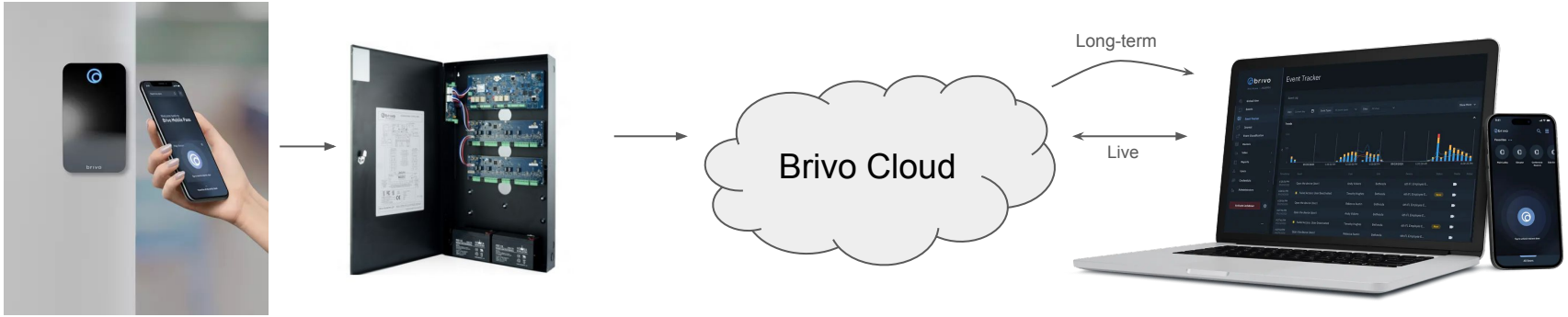
My work

- Worked on platform team (mostly backend) responsible for making SaaS platform scalable and resilient
- Interesting time to work: during rapid growth



Scaling the architecture

- Designed for 99-percentile customer (mostly small to medium businesses, office buildings, gyms, churches, etc)
- New enterprise-scale customers with massive # of sites and devices
  - Most prominent issue: **latency for live events**



GraphQL API

- Interesting problems with serverless (ex. cold start times)
- I did typical work with features requested by other teams/bug fixes

AWS AppSync

Brivo API

AWS Lambda

Internal framework

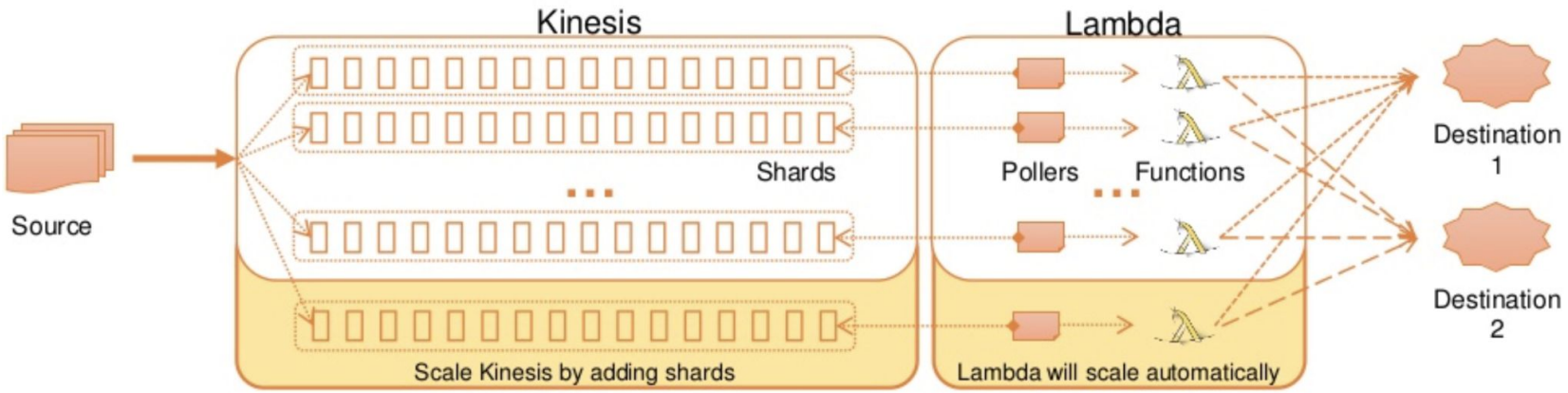
- JS utility library used by various teams/projects
- Improved Redis connections + usage for various API calls

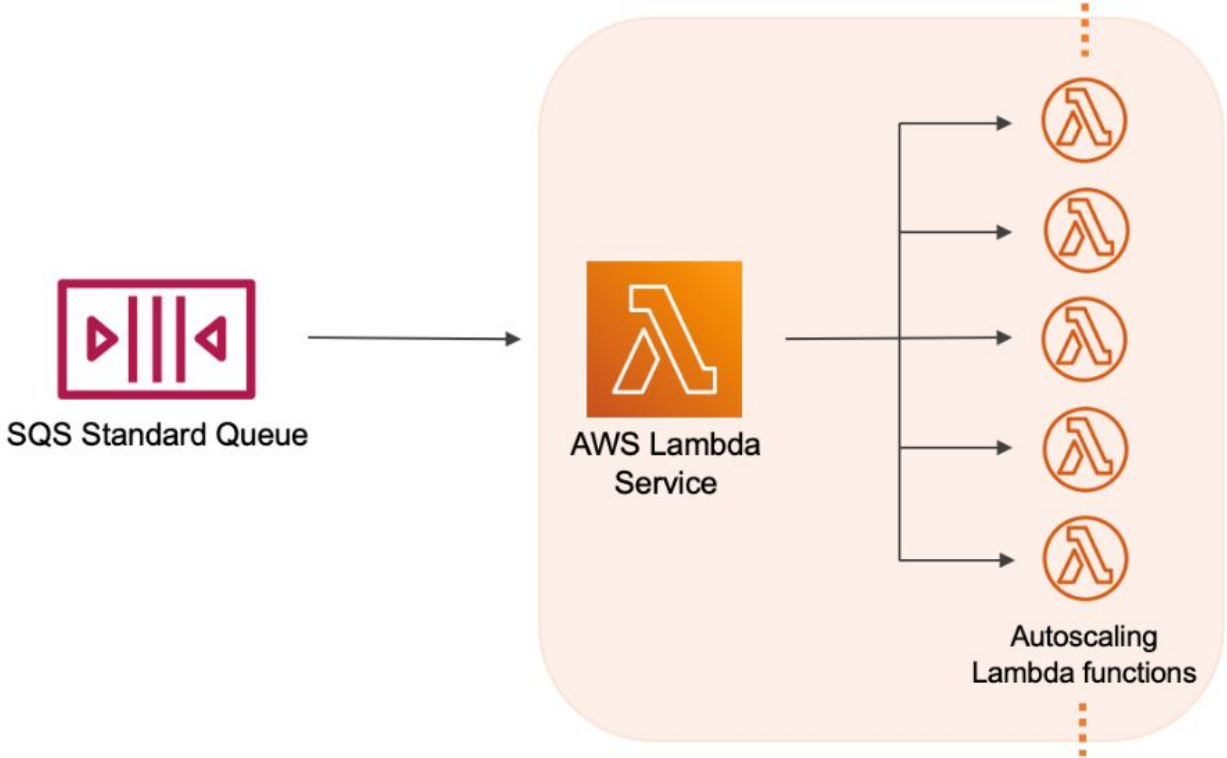
Live event processing

- Receiving events
- Real-time anomaly detection (data engineering)
- **Pushing events to WebSocket clients (front-end)**
- Pushing events to Webhook subscribers (custom back-end)

Powered by

and


Kinesis
Lambda
Source
Shards
Pollers
Functions
Destination 1
Destination 2
Scale Kinesis by adding shards
Lambda will scale automatically


SQS Standard Queue
AWS Lambda Service
Autoscaling Lambda functions